Siqi Fan

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Education

University of Chinese Academy of Sciences (UCAS) **Beijing, China**

- Master of Engineering in Automation *Sep. 2019 – Jun. 2022*
- School of Artificial Intelligence

Shanghai Jiao Tong University (SJTU) **Shanghai, China**

- Bachelor of Engineering in Automation *Sep. 2015 - Jun. 2019*
- School of Electronic Information and Electrical Engineering

Publications

- **S. Fan**, Q. Dong, F. Zhu, et al. SCF-Net: Learning spatial contextual features for large-scale point cloud segmentation. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, (**CVPR**), 2021.
- **S. Fan**, F. Zhu, S. Chen, et al. FII-CenterNet: An anchor-free detector with foreground attention for traffic object detection. In IEEE Transactions on Vehicular Technology, (**IEEE TVT**), 2021.
- **S. Fan**, F. Zhu, H. Zhang, et al. Improving road detection results based on ensemble learning and key samples focusing. In Proceedings of the IEEE International Conference on Intelligent Transportation Systems, (**ITSC**), 2020.

Research Experience

Institute for AI Industry Research, Tsinghua University (AIR) **Beijing, China**

Researcher, Computer Vision & Intelligent Vehicles *Jul. 2022 – Now*

Institute of Automation, Chinese Academy of Sciences (CASIA) **Beijing, China**

Student Researcher, Computer Vision & Intelligent Vehicles *Sep. 2019 – Jun. 2022*

- Works on the environment perception of the intelligent vehicles
  - **Introducing foreground attention to anchor-free detector for traffic object detection**
    - Proposed a foreground segmentation approach for anchor-free object detection, which could alleviate the background influences under the complex traffic environment with little extra computation cost;
    - Generated the box-induced labels of the foreground segmentation directly based on the bounding-box labels without additional labeling cost;
    - Proposed an anchor-free object detector with foreground attention;
    - Achieved the advanced performance in both accuracy and efficiency.
  - **Novel learning framework for semi-supervised semantic segmentation**
    - Proposed the conservative-progressive collaborating learning framework, which achieves the collaboration of conservative evolution and progressive exploration;

- Proposed a pseudo labeling approach based on the class-wise confusion indicator for the disagreement part of the paralleled branches;
- Proposed a confidence-based dynamic loss to reduce the influences of the noisy label;
- Achieved the SOTA performance and the advantage is more evident when fewer labeled data is leveraged. The performance of full-supervision and few-supervision is improved.

◆ **Learning spatial contextual features for large scale point cloud segmentation**

- Proposed a systematic approach for learning the spatial contextual feature, including local information representation, local and global spatial contextual feature learning;
- Designed the corresponding module and a new 3D semantic segmentation network;
- Achieved the SOTA performance for large scale point cloud semantic segmentation, and the performances on other related applications are verified by other researchers.

○ Engineering projects: mapping and localization based on LiDAR, waypoints following based on multi-sensors, lane following based on camera, object detection based on the fusion of multi-modal information (HAOMO), data interchanging based on V2X (HUAWEI).

Intel Labs China (ILC) **Beijing, China**

Autonomous System Research Intern, Intelligent Vehicles *Aug. 2020 – Dec. 2021*

○ Worked on the safety research of the intelligent vehicles

◆ **Simulation evaluation of the Responsibility -Sensitive Safety (RSS)**

- Evaluation of RSS on CARLA: build the test system, generate the test scenes and scenarios, conduct the experiments, analyze the experimental results, write the evaluation reports
- Evaluation of RSS on MATLAB: refine the test system, conduct the experiments, analyze the experimental results, write the evaluation reports

◆ **Field test of the Responsibility -Sensitive Safety (RSS)**

- Realized the test system: RTK-based perception, TCP-based transmission, CAN-based control

◆ **Cross-team cooperation and academic communication with the partner universities**

◆ **Participated in drafting the China ITS Industry Alliance Group Standard**
"Technical Requirement of Safety Assurance of AV Decision Making"

◆ The participated program was awarded **Intel's highest honor "Intel Achievement Awards"**

Intelligent Vehicle Lab, Shanghai Jiao Tong University **Shanghai, China**

Student Researcher, Computer Vision & Intelligent Vehicles *Jul. 2017 – Jun. 2019*

○ Worked on the object detection of the intelligent vehicles

◆ **Obstacle detection system for LiDAR based on embedded device**

- Initiated and led the "National Innovation Program for College Students".

◆ **3D object detection based on the fusion of multi-modal information**

Honors and Awards

○ **National Scholarship**, 2021
○ **Shanghai Jiao Tong University Excellent Scholarship**, 2018
○ **China Industrial Intelligence Challenge, Outstanding Award (State Level)**, 2018

Academic Services

○ **Reviewer of**
IEEE Transactions on Vehicular Technology, IEEE Transactions on Intelligent Vehicles, Pattern Recognition, IET Computer Vision, IET Cyber-Systems and Robotics,

IEEE Intelligent Transportation Systems Magazine,
IEEE International Conference on Intelligent Transportation Systems (2022)